GlandersWatch

Risk prediction, cited explanation and owner alerting for a neglected zoonosis

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The problem

A district officer needs to know where glanders is likely to appear next

- Glanders is a bacterial zoonosis of equids. Notifiable, often fatal, no environmental reservoir.
- An equid is a horse, donkey (recorded as asses) or mule.
- The economic weight sits in low and middle income countries, where equids are working animals.
- Surveillance reports what has already happened. No forecasting system exists for this disease.

5.50 lakh
equids in India, 2019

728
seropositive, 2019-23

138
districts affected

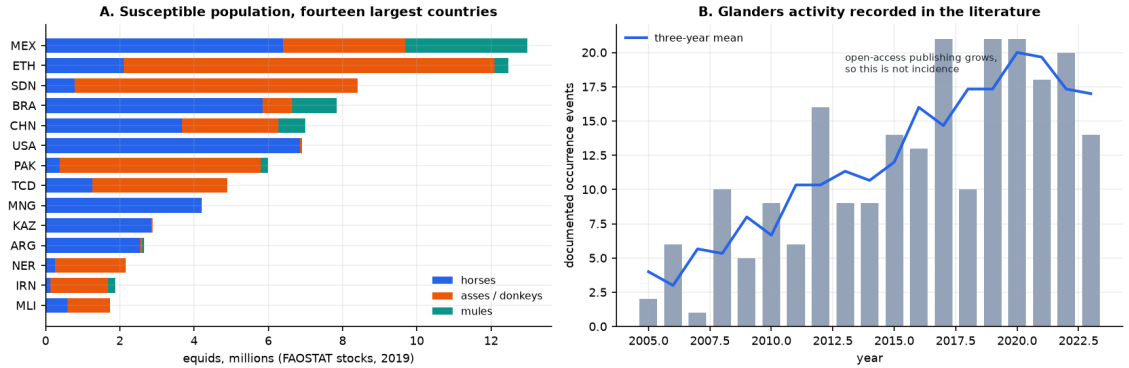
Population at risk is not the same as damage

The two quantities are routinely conflated, so they are separated at the outset

5.50 lakh equids is the size of the susceptible population: how many animals are exposed.

728 seropositive out of 158,798 screened is the documented damage: under 1% of that population.

- Because the disease is notifiable and frequently fatal once clinical, each positive is acted on rather than monitored.



Panel A: the exposed population, by species. Panel B: documented events per year, which tracks open-access publishing as much as disease. That is why the label is treated as documented activity.

What the platform is

Three components, fixed contracts, none hardcodes the disease

Component	Does	Status
Risk engine	Country features → calibrated probability, with attribution	Built and evaluated
Explanation layer	Prediction → cited answer grounded in official guidance; refuses when unsupported	Specified
Alert service	Probability → few enough messages that they are read	Specified

- A prediction alone is not usable: an owner cannot act on a probability of 0.24, and an officer cannot defend a forecast whose basis is hidden.
- A warning that arrives for every area every month teaches its recipients to ignore it.

Data

Four public sources, none previously applied to this disease

Source	Role
Europe PMC	Occurrence events from abstracts
FAOSTAT	Animal stocks, head counts
World Bank	Development indicators
NASA POWER	Climate, AG community

- 6,700 country-year rows across 268 countries assemble into the panel.
- Filtered to 2,880 modelling rows over 160 equine-holding countries, 2006–2023.
- 169 positives, a prevalence of 5.87%.
- The 3,078→2,880 change came from dropping rows with no climate-anomaly baseline rather than imputing them.

Eight features, chosen by attribution

Ranked by mean absolute SHAP on the training years only

#	Feature	Unit	Mean |SHAP|
1	pop_total	persons	1.035
2	years_since_last	years (99 = none)	0.648
3	temp_mean_anom	°C from baseline	0.505
4	internet_pct	% of population	0.449
5	humidity_anom	% RH from baseline	0.382
6	precip_cv	ratio, sd/mean	0.371
7	log_livestock	ln(1 + LSU)	0.364
8	goats	head	0.362

- 8 of 42 candidates, carrying 42% of total attribution.
- Ranking uses the training years only, so held-out years cannot influence the choice.
- Correlation and mutual information rank longitude first; attribution ranks it near the bottom. Selecting on association would keep proxies.

Model comparison

One target, two validation splits

Model	PR-AUC (temporal)	ROC-AUC (temporal)	PR-AUC (country out)
Logistic Regression	0.461	0.844	0.381
Decision Tree	0.376	0.871	0.243
Random Forest	0.495	0.874	0.322
Extra Trees	0.494	0.866	0.353
XGBoost	0.429	0.846	0.268
LightGBM	0.471	0.846	0.280

- Base rate 5.87%, so PR-AUC is the metric that moves.
- ROC-AUC spans only 0.80–0.87 across all six models on both splits: it cannot separate them.

The ranking inverts between splits

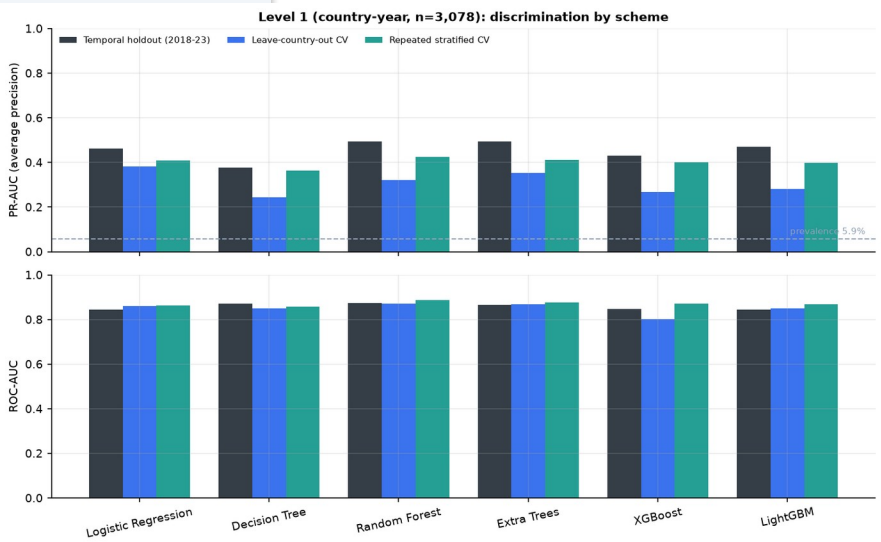
Which is the finding, not a nuisance

0.495
best on unseen years — random forest

0.381
best on unseen countries — logistic

0.322
random forest, unseen countries

- With 8 features the relationships are close enough to additive that a regularised linear model transfers best.
- The flexible ensembles pay for their extra capacity in exactly the setting the project cares about.
- Deployment means an unseen country rather than an unseen year, so the champion is logistic regression.



Calibration and alert economics

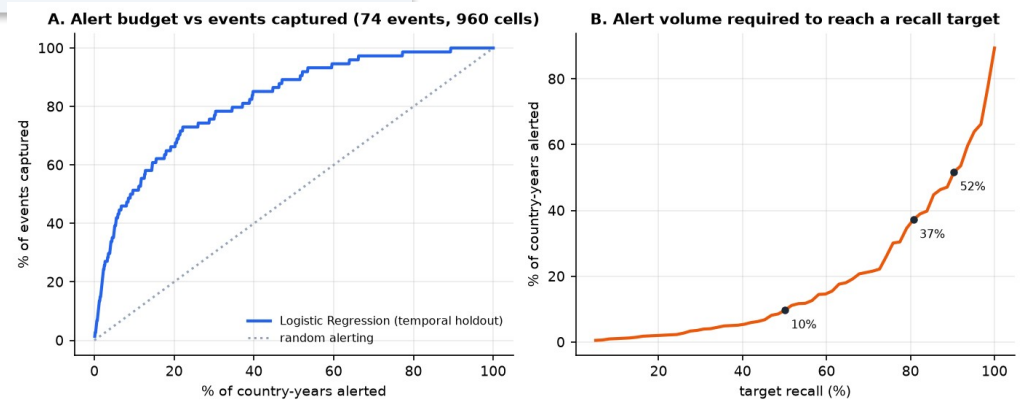
A message quoting a probability must be numerically trustworthy

0.025
expected calibration error

0.056
Brier score

43%
events caught, alerting 6%

- Half of all events are reached at an alert budget of 9% of country-years; 80% at 37%.
- Uniform alerting would need full coverage for the same recall.



The coarser question is easier

Asking whether a country is a glanders country at all

Model	ROC-AUC	PR-AUC
Logistic regression	0.951	0.914
Random forest	0.968	0.963

- 161 countries, 56 positives, prevalence 34.8%.
- Discrimination rises to 0.97 — but this is an easier question, not a better model.
- It largely asks whether a country has a reference laboratory, publishes, and has equids.
- The year-by-year target was hard precisely because reporting status is stable while incidence is not.

A negative result we kept

Predicting the calendar year does not work

Target	Model	R^2
Year of report	Linear	-0.103
Year of report	Random forest	-0.095
Year of report	Predict the mean	0.000

- Negative R^2 means worse than always predicting the average.
- The year of a report tracks when reporting began, not biology: 65% of events fall in the final five years of the window.
- A time-to-event variant scores 0.85, but only because the feature means include post-event years.

Limitations and what comes next

Limitations

- The label is documented activity, not incidence: a negative may mean no disease or no report.
- The extractor has never been checked against a hand-annotated set.
- With 169 positives the uncertainty on every headline metric is wide.
- The explanation and alert layers are specified, not deployed, so neither reports measured performance.

Next

- Model surveillance capacity as a quantity rather than a covariate.
- Validate the label extractor on a stratified sample.
- Measure the explanation layer: recall@5, faithfulness, citation precision, refusal rates.
- Pilot the alert service with a district office: messages per subscriber per year, delivery reliability.